

Fine-Tuning in Academia

Compute Challenges of RL

- Our simple RL training consumed ~**160** gpu hours (all studies ~**240 GPU days**)
- Extending performance **purely with outcome RL** not feasible in academia

Promising Directions for Fine-Tuning in Academia

- Domain specific SFT avoids computational costs
- Improve reward functions, agentic loops, custom tool integration
- New academic compute coming: Argonne's Solstice system with **100,000 NVIDIA Blackwell GPUs**

Takeaways

Fine-Tuning

- RL and SFT on synthetic QFT both **broadly increase physics performance** across all metrics
- **SFT outperforms on in-distribution** synthetic QFT
- **RL generalizes better** to QFT pedagogy, arXiv, and TP-Bench

CoT Analysis

- SFT & RL both drastically reduce errors, with **factual errors most reduced**
- **Mathematical errors remain prevalent**